

Time allowed : 3 hours

Maximum Marks : 80

GENERAL INSTRUCTIONS:

Read the following instructions very carefully and strictly follow them:

- (i) This question paper comprises **39** questions. **All** questions are compulsory.
- (ii) This question paper is divided into **five** sections- **A, B, C, D** and **E**.
- (iii) **Section A** - Question Nos. **1** to **20** are multiple choice type questions. Each question carries **1** mark.
- (iv) **Section B** - Question Nos. **21** to **26** are very short answer type questions. Each question carries **2** marks. Answer to these questions should be in the range of **30** to **50** words.
- (v) **Section C** - Question Nos. **27** to **33** are short answer type questions. Each question carries **3** marks. Answer to these questions should be in the range of **50** to **80** words.
- (vi) **Section D** - Question Nos. **34** to **36** are long answer type questions. Each question carries **5** marks. Answer to these questions should be in the range of **80** to **120** words.
- (vii) **Section E** - Question Nos. **37** to **39** are of **3** source-based/case-based units of assessment carrying **4** marks each with sub-parts.
- (viii) There is no overall choice. However, an internal choice has been provided in some sections. Only one of the alternatives has to be attempted in such questions.

SECTION-A

Select and write the most appropriate option out of the four options given for each of the questions no. 1 to 20. There is no negative marking for incorrect response.

1. In which one of the following situations a chemical reaction does **not** occur? **(Un) (1 Mark)**
 - (a) Milk is left open at room temperature during summer
 - (b) Grapes get fermented
 - (c) An iron nail is left exposed to humid atmosphere
 - (d) Melting of glaciers
2. In order to prepare dry hydrogen chloride gas in humid atmosphere the gas produced is passed through a guard tube (drying tube) which contains: **(Re) (1 Mark)**
 - (a) Calcium chloride
 - (b) Calcium oxide
 - (c) Calcium hydroxide
 - (d) Calcium carbonate
3. The property by virtue of which a solid material can be drawn into thin wires is called: **(Re) (1 Mark)**
 - (a) malleability
 - (b) ductility
 - (c) rigidity
 - (d) resistivity

4. Select from the following a hydrocarbon having one C—C bond and one C≡C bond: **(Un) (1 Mark)**

- (a) Benzene
- (b) Cyclohexane
- (c) Butyne
- (d) Propyne

5. The essential element taken up from the soil by the plants to synthesize proteins is: **(Re) (1 Mark)**

- (a) Phosphorus
- (b) Nitrogen
- (c) Iron
- (d) Magnesium

6. Select **TRUE** statements about lymph from the following: **(Re) (1 Mark)**

- A. Lymph vessels carry lymph through the body and finally open into larger arteries.
- B. Lymph contains some amount of plasma, proteins and blood cells.
- C. Lymph contains some amount of plasma, proteins and red blood cells.
- D. Lymph vessels carry lymph through the body and finally open into larger veins.

The true statements are:

- (a) A and B
- (b) B and D
- (c) A and C
- (d) C and D

7. Plants like rose and banana have lost the capacity to produce: **(Re) (1 Mark)**
 (a) flowers (b) buds
 (c) seeds (d) fruits
8. In a bisexual flower the male gametes are present in the: **(Re) (1 Mark)**
 (a) anther (b) ovary
 (c) stigma (d) filament
9. When a pure-tall pea plant is crossed with a pure-dwarf pea plant, the percentage of tall pea plants in F_1 and F_2 generation pea plants will be respectively: **(Re) (1 Mark)**
 (a) 100%, 25% (b) 100%, 50%
 (c) 100%, 75% (d) 100%, 100%
10. To get an image of magnification -1 on a screen using a lens of focal length 20 cm, the object distance must be: **(Ev) (1 Mark)**
 (a) Less than 20 cm (b) 30 cm
 (c) 40 cm (d) 80 cm
11. An optical device 'X' is placed obliquely in the path of a narrow parallel beam of light. If the emergent beam gets displaced laterally, the device 'X' is: **(Re) (1 Mark)**
 (a) plane mirror (b) convex lens
 (c) glass slab (d) glass prism
12. A piece of wire of resistance R is cut lengthwise into three identical parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R' , then the value of R/R' is: **(Ev) (1 Mark)**
 (a) $1/9$ (b) $1/3$ (c) 3 (d) 9
13. An electric bulb is rated 220 V; 11W. The resistance of its filament when it glows with a power supply of 220 V is: **(Ev) (1 Mark)**
 (a) 4400 Ω (b) 440 Ω
 (c) 400 Ω (d) 20 Ω
14. The minimum number of identical bulbs of rating 4V; 6W, that can work safely with desired brightness, when connected in series with a 240 V mains supply is: **(Ev) (1 Mark)**
 (a) 20 (b) 40 (c) 60 (d) 80
15. In the food chains given below. Select the most efficient food chain in terms of energy: **(An) (1 Mark)**
 (a) Grass $\rightarrow$ Grasshopper $\rightarrow$ Frog $\rightarrow$ Snake
 (b) Plants $\rightarrow$ Deer $\rightarrow$ Lion
 (c) Plants $\rightarrow$ Man
 (d) Phytoplankton $\rightarrow$ Zooplankton $\rightarrow$ Small Fish $\rightarrow$ Big Fish
16. Which one of the following gets biomagnified at different levels in a food chain? **(Un) (1 Mark)**
 (a) Carbon monoxide (b) CFC's
 (c) DDT (d) Manure

Question Nos. 17 to 20 consist of two statements - Assertion (A) and Reason (R). Answer these questions selecting the appropriate option (a), (b), (c) and (d) as given below:

- (a) Both, Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).
 (b) Both, Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
 (c) Assertion (A) is true, but Reason (R) is false.
 (d) Assertion (A) is false, but Reason (R) is true.
17. **Assertion (A):** In large animals, oxygen can reach different parts of the animal's body easily.
Reason (R): Respiratory pigments take up oxygen from the air and carry it to body tissues. **(Un) (1 Mark)**
18. **Assertion (A):** Concentrated nitric acid is diluted by adding water slowly to acid with constant stirring.
Reason (R): Concentrated nitric acid is easily soluble in water. **(Un) (1 Mark)**
19. **Assertion (A):** In reptiles, the temperature at which the fertilized eggs are kept decides the sex of the offsprings.
Reason (R): Sex is not genetically determined in some animals. **(Re) (1 Mark)**
20. **Assertion (A):** When ciliary muscles contract, eye lens becomes thin.
Reason (R): Ciliary muscles control the power of the eye lens. **(Un) (1 Mark)**

SECTION-B

Question Nos. 21 to 26 are very short answer type questions. Each question carries 2 marks.

21. Define oxidation. Identify and name the substance oxidised in the following reaction: **(Un) (2 Marks)**

$$\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$$
22. (A) Show the formation of magnesium chloride by electron transfer. Write the name of the cation and anion present in the compound formed. (Atomic Number of Mg = 12, Cl = 17) **(Un) (2 Marks)**
- OR**
- (B) How is zinc extracted from its ore? Name the processes involved in the extraction and write chemical equations for the reactions that occur during these processes. **(Un) (2 Marks)**
23. "Plants use a variety of techniques to get rid of waste material." Justify this statement giving any four ways. **(Ev) (2 Marks)**
24. Explain with the help of a flow chart that in human beings father is responsible for the sex (male or female) of the child. **(Cr) (2 Marks)**

25. (A) Draw a ray diagram to show the refraction of a ray of light passing through an equilateral glass prism. Mark the angle through which the emergent ray bends from the direction of the incident ray and also name it.

(Cr) (2 Marks)

OR

- (B) Name the type of lenses required by the persons for the correction of their defect of vision called presbyopia. Write the structure of the lenses commonly used for the correction of this defect giving reason for such designs.

(Re) (2 Marks)

26. What are magnetic field lines. List two important properties of magnetic field lines.

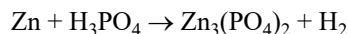
(Re) (2 Marks)

SECTION-C

Question Nos. 27 to 33 are short answer type questions. Each question carries 3 marks.

27. (A) Why do we balance a chemical equation? Name and state the law that suggests the balancing of a chemical equation? Balance the following chemical equation:

(Un) (3 Marks)



OR

- (B) Define a precipitation reaction. Give its example and also express the reaction that occurs in the form of a balanced chemical equation.
28. Design an activity to show that metals are good conductors of heat and have high melting points.
29. The digestion of food in human alimentary canal is a complex process. State the enzyme/salt present in the following and mention their function in the process of digestion :
- Saliva
 - Bile Juice
 - Pancreatic Juice
30. State two limitations of electrical impulses in multicellular organisms. Why is chemical communication better than electrical impulses as a means of communication between cells in multicellular organisms?
31. If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer.

(Ev) (3 Marks)

32. The electrical resistivity of three materials A, B and C at 20°C is given below :

(Un) (3 Marks)

Material	Resistivity ($\Omega \text{ m}$)
A	10^{17}
B	44×10^{-6}
C	1.62×10^{-8}

- Classify these materials as conductor, alloy and insulator.
 - Give one example of each of these materials and state one use of each material in the design of an electrical appliance say an electric stove or an electric iron.
33. What are decomposers? Give two examples. State how they maintain a balance in an ecosystem.

(Re) (3 Marks)

SECTION-D

Question Nos. 34 to 36 are long answer type questions. Each question carries 5 marks.

34. (A) A carbon compound 'A' on heating with excess conc. H_2SO_4 forms a compound 'B', which on addition of one mole of hydrogen gas in the presence of nickel catalyst forms a compound 'C'. 'C' on combustion in air forms 2 moles of carbon dioxide and 3 moles of water. Identify 'A', 'B' and 'C' and write their structures. Give chemical equations of the reactions involved. Also state the role of concentrated sulphuric acid in the formation of 'B' from 'A'.

(Ap) (5 Marks)

OR

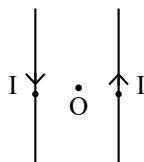
- (B) A carbon compound 'A' is widely used as a preservative in pickles and has a molecular formula $\text{C}_2\text{H}_4\text{O}_2$. This compound reacts with ethanol to form a sweet smelling compound 'B'.
- Identify the compound 'A' and write its structure.
 - Write the chemical equation for the reaction of 'A' with ethanol to form compound 'B'. State the role of presence of an acid in the reaction.
 - How can we get compound 'A' back from 'B'?
 - How can 'A' be obtained from ethanol?
 - Name the gas produced when compound 'A' reacts with washing soda.
35. (A) (i) What is regeneration? Give one example of an organism that shows this process and one organism that does not. Why does regeneration not occur in the latter?
- (ii) Water in a pond appears dark green and contains filamentous structures. Name these structures and the method by which they reproduce. Explain the process.

OR

- (B) (i) Name the part performing following functions in human male reproductive system:
- Carries sperm
 - Production of male gametes
 - Whose secretion makes the transport of sperms easier
 - Provide suitable temperature for sperm formation
- (ii) Write any two characteristics of sperms.
- (iii) What are surgical contraceptive methods? Give the side effect caused by this procedure.

(Re) (5 Marks)

36. (A) (i) Draw the pattern of the magnetic field lines for the two parallel straight conductors carrying current of same magnitude I in opposite directions as shown. Show the direction of magnetic field at a point O which is equidistant from the two conductors. (Consider that the conductors are inserted normal to the plane of a rectangular cardboard.) (Cr)

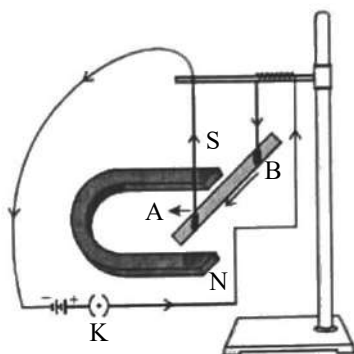


- (ii) In our houses we receive A.C. electric power of 220 V. In electric iron or electric heater cables having three wires with insulation of three different colours - red, black and green are used to draw current from the mains.
- What are these three different wires called? Name them colourwise.
 - What is the potential difference between the red wire and the black wire?
 - What is the role of the wire with green insulation in case of accidental leakage of electric current to the metallic body of an electrical appliance?

(Re) (5 Marks)

OR

- (B) (i) By using the given experimental set-up. How can it be shown that:



- a force is exerted on the current-carrying conductor AB when it is placed in a magnetic field.
 - the direction of force can be reversed in two ways.
- (ii) When will the magnitude of the force be highest?
- (iii) State Fleming's left hand rule. (Un) (5 Marks)

SECTION-E

Question Nos. 37 to 39 are Source-based/Case-based questions.

37. Common salt is a very important chemical compound for our daily life. It's chemical name is sodium chloride and it is used as a raw material in the manufacture of caustic soda, washing soda, baking soda etc. It is also used in the preservation of pickles, butter, meat etc. (Un)
- Name the acid and the base from which common salt can be obtained. (1 Mark)
 - State the nature (acidic/basic/neutral) of sodium chloride. Give reason for the justification for your answer. (1 Mark)
 - (A) What happens when electric current is passed through an aqueous solution of sodium chloride (called brine)? Name the products obtained along with the corresponding places in the electrolytic cell where each of these products is obtained. (2 Marks)

Or

- (B) How is washing soda obtained from sodium chloride? Give chemical equation of the reactions involved in the process. (2 Marks)
38. In life there are certain changes in the environment called 'stimuli to which we respond appropriately. Touching a flame suddenly is a dangerous situation for us. One way is to think consciously about the possibility of burning and then moving the hand. But our body has been designed in such a way that we save ourselves from such situations immediately.
- Name the action by which we protect ourselves in the situation mentioned above and define it. (1 Mark)
 - Write the role of (a) motor and (b) relay neuron. (1 Mark)
 - (A) What are the two types of nervous system in human body? Name the components of each of them. (2 Marks)

Or

(iii)(B) Which part of the human brain is responsible for:
(2 Marks)

- (a) thinking
- (b) picking up a pencil
- (c) controlling blood pressure
- (d) controlling hunger

39. The students in a class took a thick sheet of cardboard and made a small hole in its centre. Sunlight was allowed to fall on this small hole and they obtained a narrow beam of white light. A glass prism was taken and this white light was allowed to fall on one of its faces. The prism was turned slowly until the light that comes out of the opposite face of the prism appeared on the nearby screen. They studied this beautiful band of light and concluded that it is a spectrum of white light.
(Re, Cr)

(i) Give any one more instance in which this type of spectrum is observed.
(1 Mark)

(ii) What happens to white light in the above case?
(1 Mark)

(iii)(A) List two conditions necessary to observe a rainbow.
(2 Marks)

Or

(iii)(B) Draw a ray diagram to show the formation of a rainbow. Mark on it, points (a), (b) and (c) as given below :
(2 Marks)

- (a) Where dispersion of light occurs.
- (b) Where light gets reflected internally.
- (c) Where final refraction occurs.

EXPLANATIONS

1. (d) Melting of glaciers is an example of physical change.
[NCERT, Ch-1, Pg. no. 1, 2] (1 Mark)

2. (a) Pass the gas produced through a guard tube (drying tube) containing calcium chloride to dry the gas.
[NCERT, Ch-2, Pg. no. 23] (1 Mark)

3. (b) Ductility is the property by virtue of which a solid material can be drawn into thin wires.
(1 Mark)
[NCERT, Ch-3, Pg. no. 38]

4. (d) Propyne is $\text{CH}_3\text{—C}\equiv\text{CH}$. Hence, it contains one C—C bond and one $\text{C}\equiv\text{C}$ bond.
(1 Mark)
[NCERT, Ch-4, Pg. no. 64-68]

5. (b) Nitrogen is absorbed from the soil in the form of nitrates and ammonium ions, which are essential for protein synthesis in plants.
(1 Mark)
[NCERT, Ch-5, Pg. no. 83, 84]

6. (b) Lymph vessels do not open into arteries but into veins. It contains plasma, proteins, and blood cells, excluding red blood cells.
(1 Mark)
[NCERT, Ch-5, Pg. no. 94]

7. (c) Plants like bananas and roses reproduce through vegetative propagation as they have lost the capacity to produce seeds.
(1 Mark)
[NCERT, Ch-7, Pg. no. 117]

8. (a) The anther is part of the stamen, which produces pollen grains containing male gametes in flowering plants.
(1 Mark)
[NCERT, Ch-7, Pg. no. 120]

9. (c) In the F_1 generation, all pea plants are tall (100%) due to dominant inheritance. In the F_2 generation, the tall and dwarf plants appear in a 3:1 ratio, meaning 75% tall and 25% dwarf.
(1 Mark)
[NCERT, Ch-8, Pg. no. 130]

10. (c) For a thin lens, the magnification m is given by $m = \frac{v}{u}$. When $m = -1$, we have $v = -u$ in magnitude (the negative sign indicates the image is inverted). Using the lens formula
$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$
substitute $v = -u$ to obtain

$$\frac{1}{20} = \frac{1}{-u} - \frac{1}{u} = \frac{-u}{u}$$

Solving for u gives

$$u = -40 \text{ cm.} \quad (1 \text{ Mark}) \quad [\text{NCERT, Ch-9, Pg. no. 155}]$$

11. (c) A glass slab laterally displaces an obliquely incident beam due to refraction at its surfaces.
(1 Mark)
[NCERT, Ch-9, Pg. no. 148]

12. (d) Cutting the wire into three equal lengths gives each piece a resistance of $R/3$; connecting three such resistors in parallel yields an equivalent resistance of
$$\frac{1}{R'} = \frac{3}{R} + \frac{3}{R} + \frac{3}{R} \Rightarrow \frac{1}{R'} = \frac{3+3+3}{R}$$

$$\frac{1}{R'} = \frac{9}{R} \Rightarrow R' = \frac{R}{9}$$

$$\text{So, } \frac{R}{R'} = \frac{R}{\frac{R}{9}} = 9. \quad (1 \text{ Mark})$$

[NCERT, Ch-11, Pg. no. 186]

13. (a) Using the formula $R = \frac{V^2}{P}$
 substituting $V = 220 \text{ V}$ and $P = 11 \text{ W}$
 Gives $R = \frac{220^2}{11} = \frac{48400}{11} = 4400 \Omega$ (1 Mark)
 [NCERT, Ch-11, Pg. no. 191]

14. (c) Each bulb is designed to operate at 4 V. When connected in series, the supply voltage divides equally among all bulbs. To ensure that each bulb receives 4 V, we must have:

$$\text{Number of bulbs} = \frac{240 \text{ V}}{4 \text{ V per bulb}} = 60$$

Thus, a minimum of 60 bulbs are required for safe operation with desired brightness. (1 Mark)

[NCERT, Ch-11, Pg. no. 184]

15. (c) The shorter the food chain, the more energy is conserved. Since 10 % energy is lost at each trophic level, a short consumption food chain like Plants → Man is the most energy-efficient. (1 Mark)

[NCERT, Ch-13, Pg. no. 211]

16. (c) DDT is a non-biodegradable pesticide that accumulates in organisms at higher trophic levels through biomagnification, affecting top consumers most severely. (1 Mark) [NCERT, Ch-13, Pg. no. 212]

17. (a) In large animals, oxygen reaches different body parts efficiently due to the presence of respiratory pigments like hemoglobin, which bind to oxygen and carry it through the circulatory system. This allows oxygen to be transported to various tissues, ensuring easy distribution throughout the body. (1 Mark)

[NCERT, Ch-5, Pg. no. 90]

18. (d) Concentrated nitric acid is diluted by adding slowly to water. The acid must always be added slowly to water with constant stirring. (1 Mark)

[NCERT, Ch-2, Pg. no. 24]

19. (a) In some reptiles, the temperature at which eggs are incubated decides the offspring's sex. Since sex is not genetically determined in such cases, Reason (R) correctly explains Assertion (A). (1 Mark)

[NCERT, Ch-8, Pg. no. 132]

20. (d) (1 Mark) [NCERT, Ch-10, Pg. no. 161]

21. An oxidation reaction is a chemical reaction in which a substance gains oxygen or loses hydrogen during a reaction. (1 Mark)

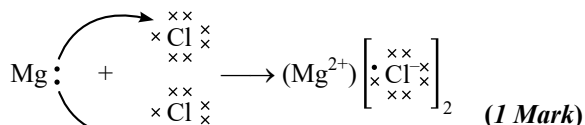
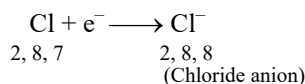
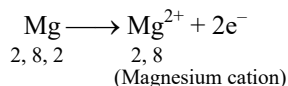
In the given reaction:

The copper(II) oxide is losing oxygen and is being reduced. (½ Mark)

The hydrogen is gaining oxygen and is being oxidised. (½ Mark)

[NCERT, Ch-1, Pg. no. 12]

22. (A) The formation of magnesium chloride by electron transfer is given below:



The name of the cation: Magnesium ion (½ Mark)

The name of the anion: Chloride ion (½ Mark)

[NCERT, Ch-3, Pg. no. 48]

OR

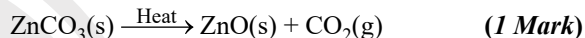
- (B) Zinc can be extracted from its ore as follows:-

First, Zinc sulphide or Zinc carbonate is converted into zinc oxide either by roasting or calcination.

Roasting:

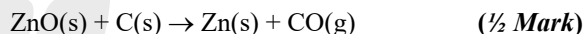


Calcination:



The zinc oxides are then reduced to the corresponding metals by using suitable reducing agents such as carbon. (½ Mark)

For example, when zinc oxide is heated with carbon, it is reduced to metallic zinc.

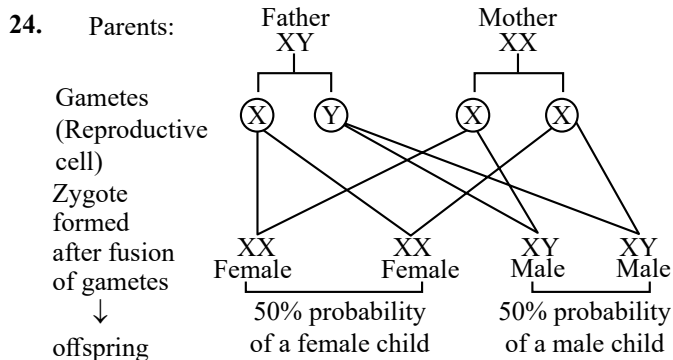


[NCERT, Ch-3, Pg. no. 51]

23. Plants eliminate waste using different methods:

- Oxygen Removal: Plants release excess oxygen produced during photosynthesis. (½ Mark)
- Transpiration: Excess water is removed through transpiration. (½ Mark)
- Storage in Tissues: Wastes are stored in vacuoles, fallen leaves, resins, and gums. (½ Mark)
- Excretion into Soil: Some waste substances are released into the surrounding soil. (½ Mark)

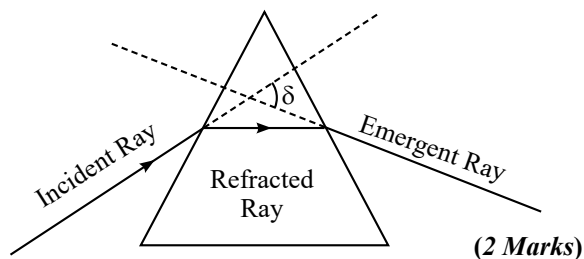
[NCERT, Ch-5, Pg. no. 98]



Since the mother always contributes an X chromosome, the father's sperm (X or Y) determines whether the child will be female (XX) or male (XY). (2 Marks)

[NCERT, Ch-8, Pg. no. 132]

25. (A) The ray diagram to show the refraction of a ray of light passing through an equilateral glass prism



OR

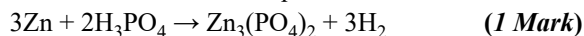
- (B) To correct this defect a presbyopia a bifocal lens made by a combination of the concave lens (upper part) and convex lens (lower part) is used because they combine a concave lens for distance vision with a convex lens for near vision. (2 Marks)
26. Magnetic field lines are imaginary lines that represent the direction and strength of a magnetic field, always originating from the North Pole of a magnet and terminating at the South Pole. They have two key properties:
1. They form continuous closed loops.
 2. They never intersect.

(2 Marks)

27. (A) A chemical equation is balanced to ensure that the number of atoms of each element remains the same, before and after a chemical reaction. (1 Mark)
- This is necessary to follow the Law of Conservation of Mass, which states that mass can neither be created nor destroyed in a chemical reaction; the total mass of reactants is equal to the total mass of products.

(1 Mark)

The balanced chemical equation is:



(1 Mark)

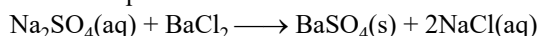
[NCERT, Ch-1, Pg. no. 3-5]

OR

- (B) Any reaction that produces a precipitate can be called a precipitation reaction. (1 Mark)

For example: The white precipitate of BaSO_4 is formed by the reaction of sodium sulphate with barium chloride. (1 Mark)

The complete balanced equation for the reaction of sodium sulphate with barium chloride is

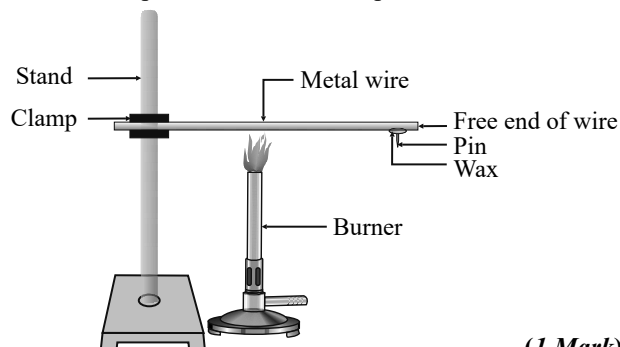


(Sodium sulphate) (Barium chloride) (Barium sulphate) (Sodium chloride)

(1 Mark)

[NCERT, Ch-1, Pg. no. 11, 12]

28. 1. Take an aluminium or copper wire.
2. Clamp this wire on a stand, as shown in Fig.
3. Fix a pin to the free end of the wire using wax.
4. Heat the wire with a spirit lamp, candle or a burner near the place where it is clamped.



(1 Mark)

Observations:

1. When the metal wire (aluminium or copper) is heated near the clamped end, heat starts travelling through the metal by conduction, and the pin fixed to the free end of the wire using wax will eventually fall off due to the heat conducted along the wire, melting the wax. (½ Mark)
2. The metal wire itself does not melt, because aluminium and copper have high melting points. (½ Mark)

Conclusion:

- Metals are good conductors of heat as heat travels from the heated end to the other end. (½ Mark)
- Since the metal wire does not melt, it proves that aluminium and copper can withstand high temperatures and have high melting points. (½ Mark)

The above activity shows that metals are good conductors of heat and have high melting points.

[NCERT, Ch-3, Pg. no. 38]

29. (i) Saliva contains enzyme salivary amylase, which starts starch digestion in the mouth. (1 Mark)
- (ii) Bile juice, secreted by the liver, contains bile salts, which emulsify fats, increasing enzyme efficiency. (1 Mark)
- (iii) Pancreatic juice from the pancreas contains trypsin (help in protein digestion) and lipase (for breaking down emulsified fats). (1 Mark)

[NCERT, Ch-5, Pg. no. 85, 86]

30. Limitations of Electrical Impulses:

- Limited Reach: Electrical impulses can only reach cells connected through nervous tissue and cannot communicate with each and every cell in the animal body. (1 Mark)
- Short-term Effect: Once generated, the cell needs time to reset before transmitting another impulse, preventing continuous signal transmission. (1 Mark)

- Chemical communication is better than electrical communication because chemical signals (hormones) can reach all cells regardless of nervous connections and ensure a long-lasting and steady response, making coordination more effective. (1 Mark)

[NCERT, Ch-6, Pg. no. 108]

31. For a concave mirror with a focal length of 18 cm, a virtual and magnified image is obtained when the object is placed between the mirror and its focus. Using the mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

and the magnification formula

$$m = \frac{-v}{u}$$

given $m = +2$, we have $v = -2u$. Substituting in the mirror formula:

$$\frac{1}{18} = \frac{1}{-2u} + \frac{1}{u} = \frac{1}{2u}$$

which gives $u = \frac{18}{2} = 9$ cm. Thus, placing the object 9 cm

from the mirror produces a virtual, erect, and magnified image. [NCERT, Ch-9, Pg. no. 143]

32. (i) A → Insulator (highest resistivity) (½ Mark)
 B → Alloy (moderate resistivity) (½ Mark)
 C → Conductor (lowest resistivity) (½ Mark)

(ii) **Conductor (e.g., Copper)**

Use: Copper wiring in an electric iron or stove for efficient current flow. (½ Mark)

Alloy (e.g., Nichrome)

Use: Heating element in electric iron or stove due to its relatively higher resistivity and ability to withstand high temperatures. (½ Mark)

Insulator (e.g., Plastic or Rubber)

Use: Insulating handle or covering to prevent electric shocks. (½ Mark) [NCERT, Ch-11, Pg. no. 179]

33. Decomposers are microorganisms that break down complex organic matter into simple inorganic substances, recycling nutrients in the ecosystem. (1 Mark)

Examples:

- Bacteria (½ Mark)
- Fungi (½ Mark)

Role in Maintaining Ecosystem Balance:

- Nutrient Recycling: Convert dead matter into nutrients for plants, maintaining soil fertility. (½ Mark)
- Waste Management: Decompose organic waste, preventing accumulation and environmental pollution. (½ Mark)

[NCERT, Ch-13, Pg. no. 209]

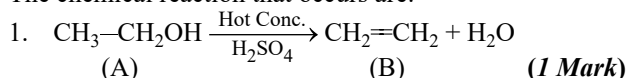
34. (A) A is Ethanol- $\text{CH}_3\text{—CH}_2\text{—OH}$

B is Ethene- $\text{CH}_2=\text{CH}_2$

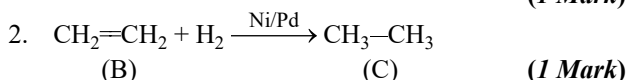
C is Ethane- $\text{CH}_3\text{—CH}_3$

(1 Mark)

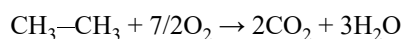
The chemical reaction that occurs are:



- The concentrated sulphuric acid can be regarded as a dehydrating agent which removes water from ethanol. (1 Mark)



3. The balanced chemical reaction for the combustion of ethane is:



(C)

(1 Mark)

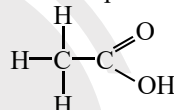
[NCERT, Ch-4, Pg. no. 69-72]

OR

- (B) The molecular formula $\text{C}_2\text{H}_4\text{O}_2$ corresponds to ethanoic acid (acetic acid, CH_3COOH).

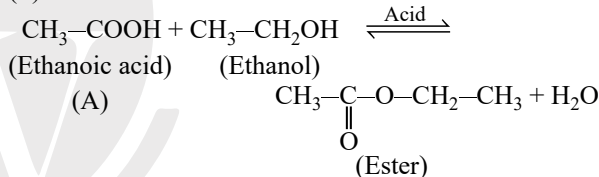
It is widely used as a preservative in pickles.

- (i) The compound A is Ethanoic acid and its structure is



(1 Mark)

- (ii) The reaction of A with ethanol is shown below:

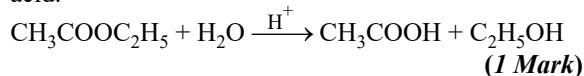


(½ Mark)

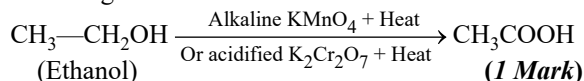
Role of Acid (H_2SO_4):

- Concentrated sulphuric acid (H_2SO_4) acts as a catalyst and dehydrating agent, which helps in esterification. (½ Mark)

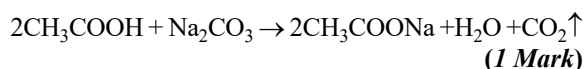
- (iii) Hydrolysis of ester (ethyl ethanoate) in the presence of a dilute acid (HCl or H_2SO_4) regenerates ethanoic acid.



- (iv) Compound A can be obtained from Ethanol by the following reaction:



- (v) Washing soda is $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$. The reaction that occurs is



[NCERT, Ch-4, Pg. no. 71-74]

35. (A) (i)

- Regeneration is the ability of fully differentiated organisms to give rise to new individuals from their body parts using specialized cells. (1 Mark)
- *Planaria* shows regeneration, if cut into pieces, each piece can develop into a complete organism. (½ Mark)
- Humans are highly differentiated organisms, they do not show regeneration. (½ Mark)
- In complex organisms like humans, cells are specialized into tissues and organs with distinct functions. This specialization and lack of regenerative cells prevents complete regeneration of the body from a single part. (1 Mark)

(ii)

The dark green filamentous structures in the pond can be *Spirogyra*, a type of green algae. (½ Mark)

Spirogyra reproduces asexually by fragmentation. (½ Mark)

Process of fragmentation:

- It occurs when a mature *Spirogyra* filament breaks into smaller pieces. (½ Mark)
- Each fragment has the ability to regenerate and form a fully functional filament. (½ Mark)

[NCERT, Ch-7, Pg. no. 116, 117]

OR

(B) (i) Carries sperm – Vas deferens (½ Mark)

Production of male gametes – Testes (½ Mark)

Secretion making sperm transport easier – Seminal vesicles and prostate gland (½ Mark)

Provides suitable temperature for sperm formation – Scrotum (½ Mark)

(ii) Characteristics of sperms:

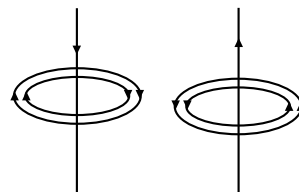
- Motile – Sperms have a tail that helps them swim towards the female gamete. (½ Mark)
- Haploid – They carry half the genetic material (23 chromosomes) for fertilization. (½ Mark)

(iii) Surgical contraceptive methods are permanent birth control methods where, in males, sperm transport is blocked by cutting the vas deferens, and in females, egg movement is prevented by cutting the fallopian tubes. (1 Mark)

Side Effect: These procedures, though safe but can cause post-surgical complications or infections if not performed correctly. (1 Mark)

[NCERT, Ch-7, Pg. no. 123, 125]

36. (A) (i) By applying the right-hand thumb rule, we can sketch the magnetic field lines around the given wires. Since first wire carry current downward, so it's magnetic field lines run clockwise, and second wire carry current upward, so it's magnetic field lines run anticlockwise, as shown in the figure. (1 Mark) [NCERT, Ch-12, Pg. no. 198]



(1 Mark)

(ii) (a) Red wire: Live (or Phase)

Black wire: Neutral

Green wire: Earth (or Ground) (1 Mark)

(b) The potential difference between the red (live) and black (neutral) wires is 220 V (in India). (1 Mark)

(c) The green (earth) wire provides a direct path to ground in case of accidental current leakage to the metallic body of an appliance, preventing electric shock by allowing the current to safely flow to earth instead of through a person. (1 Mark)

OR

(B) (i) Demonstration with Experimental Set-up

(a) Force on Conductor:

- Arrange the conductor AB so that it is placed between the poles of a U-shaped magnet (with North and South poles facing each other).
- Connect AB to a battery or power source so that current flows through the conductor.
- On switching the current ON, the conductor is observed to move or jerk in a particular direction. This shows that a current-carrying conductor experiences a force when placed in a magnetic field.

(b) Reversing Direction of Force:

- **Reverse the direction of current:** Swap the battery connections so the current flows in the opposite direction. You will see that the conductor moves in the opposite direction.
- **Reverse the polarity of the magnet:** Flip the magnet so that the North and South poles are interchanged. The direction of force on the conductor reverses again.

(ii) Condition for Maximum Force

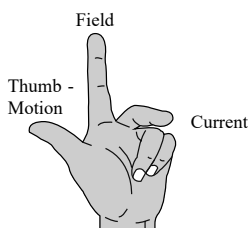
The force on a current-carrying conductor in a magnetic field is given by

$$F = B I l \sin \theta$$

The force is **maximum** when $\theta = 90^\circ$, i.e., the conductor is placed **perpendicular** to the magnetic field lines

(iii) Fleming's Left Hand Rule

If you stretch the **thumb**, **forefinger**, and **middle finger** of your left hand mutually at right angles to each other, the **Forefinger (index finger)** points in the direction of the **Field** (N to S), the **Middle finger** points in the direction of the **Current** (positive to negative) and the **Thumb** then shows the direction of the **Force** (motion) on the conductor. [NCERT, Ch-12, Pg. no. 202]



37. (i) **Acid:** Hydrochloric acid (HCl) (½ Mark)

Base: Sodium hydroxide (NaOH) (½ Mark)

(ii) It is formed by the reaction of a strong acid (HCl) and a strong base (NaOH), which results in a neutral salt. (1 Mark)

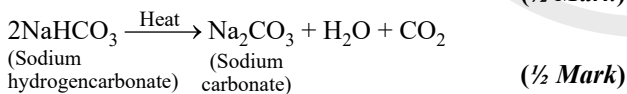
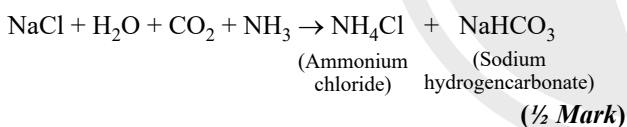
(iii) (A) When electricity is passed through an aqueous solution of sodium chloride (called brine), it decomposes to form sodium hydroxide. (½ Mark)

Electrolysis of brine (NaCl solution) produces:

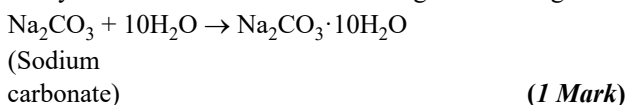
- Hydrogen gas (H₂) at the cathode. (½ Mark)
 - Chlorine gas (Cl₂) at the anode. (½ Mark)
 - Sodium hydroxide solution is formed near the cathode
- $$2\text{NaCl(aq)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{Cl}_2\text{(g)} + \text{H}_2\text{(g)}$$
- (½ Mark)

OR

(iii) (B) Washing soda is obtained from NaCl by the following equations:



Recrystallisation of sodium carbonate gives washing soda.



[NCERT, Ch-2, Pg. no. 29-31]

38. (i)

- The action is called a reflex action. (½ Mark)
- A reflex action is an automatic and rapid response to a stimulus that occurs without conscious thought to protect the body from harm. (½ Mark)

(ii) Role of:

1. Motor Neuron: It carries impulses from the

central nervous system (CNS) to the muscles, enabling movement or response. (½ Mark)

2. Relay Neuron: It transmits impulses between sensory neurons and motor neurons within the spinal cord, forming a reflex arc for quick responses. (½ Mark)

(iii) (A) Two types of the nervous system in the human body and their components are:

- Central Nervous System (CNS) – Includes the brain and spinal cord. (1 Mark)
- Peripheral Nervous System (PNS) – Includes cranial nerves and spinal nerves. (1 Mark)

OR

(iii) (B) The part of the human brain is responsible for:

- Thinking – Fore-brain (½ Mark)
- Picking up a pencil – Cerebellum (½ Mark)
- Controlling blood pressure – Medulla oblongata (½ Mark)
- Controlling hunger – Hypothalamus (½ Mark)

[NCERT, Ch-6, Pg. no. 103, 104]

39. (i) A rainbow in the sky after rainfall is a common natural example of the same type of spectrum (splitting of white light into its constituent colors). (1 Mark)

(ii) When white light passes through the prism, it undergoes refraction and dispersion. Different colors (wavelengths) of light bend by different amounts, causing the white light to split into a band of seven colors (VIBGYOR). (1 Mark)

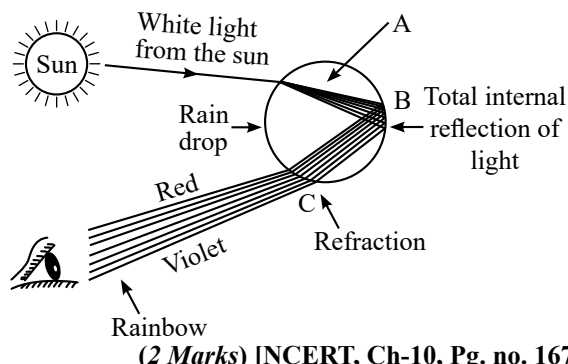
(iii) (A) Two conditions necessary to observe a rainbow

- The Sun should be behind the observer. (1 Mark)
- There must be water droplets (from rain or mist) in front of the observer. (1 Mark)

OR

(iii) (B) Ray diagram for the formation of a rainbow

Below is a simplified depiction of how a single raindrop disperses white light: (1 Mark)



2025

Outside DELHI CBSE Solved Paper

Time allowed : 3 hours

Maximum Marks : 80

GENERAL INSTRUCTIONS:

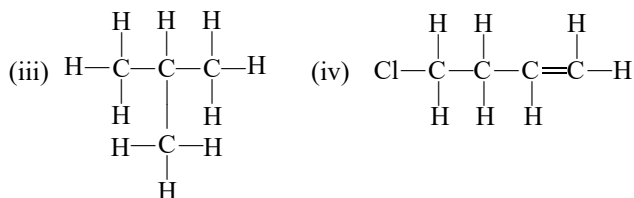
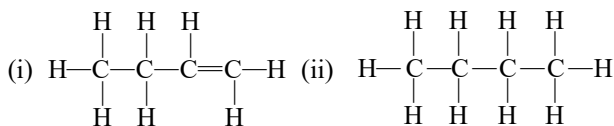
Read the following instructions very carefully and strictly follow them:

- (i) This question paper comprises **39** questions. **All** questions are compulsory.
- (ii) This question paper is divided into **five** sections- **A, B, C, D** and **E**.
- (iii) **Section A** - Question Nos. **1** to **20** are multiple choice questions. Each question carries **1** mark.
- (iv) **Section B** - Question Nos. **21** to **26** are very short answer type questions. Each question carries **2** marks. Answer to these questions should be in the range of **30** to **50** words.
- (v) **Section C** - Question Nos. **27** to **33** are short answer type questions. Each question carries **3** marks. Answer to these questions should be in the range of **50** to **80** words.
- (vi) **Section D** - Question Nos. **34** to **36** are long answer type questions. Each question carries **5** marks. Answer to these questions should be in the range of **80** to **120** words.
- (vii) **Section E** - Question Nos. **37** to **39** are of **3** source-based/case-based units of assessment carrying **4** marks each with sub-parts.
- (viii) There is no overall choice. However, an internal choice has been provided in some sections. Only one of the alternatives has to be attempted in such questions.

SECTION-A

Select and write the most appropriate option out of the four options given for each of the questions no. 1 to 20. There is no negative marking for incorrect response.

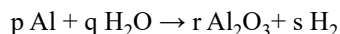
1. Given below are the structures of some hydrocarbons. Select the two structures which are related to each other from the given options: (An) (1 Mark)



- (a) (i) and (iv) (b) (ii) and (iv)
(c) (ii) and (iii) (d) (i) and (iii)
2. Choose the **incorrect** statement about the common reaction used in hydrogenation of vegetable oils. (Un) (1 Mark)

- (a) It is an addition reaction.
(b) It takes place in the presence of nickel or palladium catalyst.
(c) The product contains only single bonds between carbon atoms.
(d) It is an addition reaction which occurs in the presence of an acid catalyst.

3. Consider the following chemical equation: (Ev) (1 Mark)



To balance this chemical equation, the values of 'p', 'q', 'r' and 's' must be respectively:

- (a) 3,2,2,1 (b) 2,3,3,1 (c) 2,3,1,3 (d) 3,1,2,2

4. Study the following cases: (1 Mark)

- (i) $\text{CuSO}_4 + \text{Mg} \rightarrow$
(ii) $\text{FeSO}_4 + \text{Pb} \rightarrow$
(iii) $\text{CaSO}_4 + \text{Al} \rightarrow$
(iv) $\text{ZnSO}_4 + \text{Ca} \rightarrow$

The case/cases in which new product (s) will form is/are: (Un)

- (a) Only (i) (b) Only (iii)
(c) (i) and (iv) (d) (i), (ii) and (iv)

5. Which of the given option represents a family of salts?

(1 Mark)(Re)

- (a) NaCl, Na₂SO₄, CaSO₄
- (b) K₂SO₄, Na₂SO₄, CaSO₄
- (c) NaNO₃, CaCO₃, Na₂CO₃
- (d) MgSO₄, CuSO₄, MgCl₂

6. Match Column-I with Column-II and select the correct option from the choices provided. (Re) (1 Mark)

Column-I		Column-II	
A.	Site of fertilisation of egg with the sperm	(i)	Vagina
B.	Site of implantation of embryo	(ii)	Uterus
C.	Site of entry of sperm into the female reproductive tract	(iii)	Oviduct
D.	Site through which the waste materials generated by the developing embryo are removed	(iv)	Placenta
		(v)	Cervix

- (a) A -(ii), B -(i), C -(v), D -(iv)
- (b) A -(iii), B -(i), C -(v), D -(iv)
- (c) A -(iv), B -(ii), C -(iii), D -(i)
- (d) A -(iii), B -(ii), C -(i), D -(iv)

7. The part of the hind-brain controlling involuntary actions such as salivation and vomiting in humans is: (Re) (1 Mark)

- (a) Cerebellum
- (b) Cerebrum
- (c) Pons
- (d) Medulla

8. Select a pair of bisexual flowers from the following : (Re) (1 Mark)

- (a) Papaya and mustard
- (b) Hibiscus and mustard
- (c) Hibiscus and papaya
- (d) Hibiscus and watermelon

9. The plant hormone present in greater concentration in the areas of rapidly dividing cells is: (Re) (1 Mark)

- (a) Auxin
- (b) Cytokinins
- (c) Gibberellins
- (d) Absciscic acid

10. Temporary finger-like extensions on the cell surface to take in food is formed in: (Re) (1 Mark)

- (a) *Paramoecium*
- (b) *Amoeba*
- (c) *Leishmania*
- (d) *Rhizopus*

11. A wire of length 'l' is gradually stretched so that its length increases to 3l. If its original resistance is R, then its new resistance will be: (Ev) (1 Mark)

- (a) 3R
- (b) 6R
- (c) 9R
- (d) 27R

12. Which one of the following statements is **not** true about a bar magnet ? (Re) (1 Mark)

- (a) It sets itself in north-south direction when suspended freely.
- (b) It has attractive power for iron filings.
- (c) It produces magnetic field lines.
- (d) The direction of magnetic field lines inside a bar magnet is from its north pole to its south pole.

13. Absolute refractive index of water and glass is $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is: (Ev) (1 Mark)

- (a) $\frac{9}{4}$ m/s
- (b) $\frac{7}{3}$ m/s
- (c) $\frac{16}{9}$ m/s
- (d) $\frac{9}{8}$ m/s

14. When a beam of white light passes through a region of very fine dust particles, the colour of light that scatters the most in that region is : (Re) (1 Mark)

- (a) red
- (b) orange
- (c) blue
- (d) yellow

15. The strength of the magnetic field produced inside a current carrying long straight solenoid does not depend upon : (Un) (1 Mark)

- (a) Area of cross-section of the tube of the solenoid
- (b) Number of turns per unit length in the solenoid
- (c) Direction of current flowing through the coil of the solenoid
- (d) Material of the core of the solenoid

16. Other than the abiotic components, which of the given biotic components are **not** required to make an aquarium with small herbivorous fishes a self-sustaining system? (Un) (1 Mark)

- (i) Aquatic plants and aquatic animals
- (ii) Terrestrial plants and terrestrial animals
- (iii) Decomposers as bacteria and fungi
- (iv) Consumers as clown fishes and sea urchins
- (a) (i) and (iv)
- (b) (ii) and (iii)
- (c) (i) and (iii)
- (d) (ii) and (iv)

For Questions number 17 to 20, two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of Assertion (A).
- (c) Assertion (A) is true, but Reason (R) is false.
- (d) Assertion (A) is false, but Reason (R) is true.

17. Assertion (A): In our actions of writing or talking, our nervous system communicates with the muscles.

Reason (R): Cranial nerves and spinal nerves form the peripheral nervous system. **(Re) (1 Mark)**

18. Assertion (A): Use of jute bags for shopping reduces pollution.

Reason (R): Jute is biodegradable and its bag may be reused as and when needed. **(Un) (1 Mark)**

19. Assertion (A): Hydrogen gas is not evolved when a metal reacts with nitric acid.

Reason (R): Nitric acid is a strong reducing agent and reduces the hydrogen produced in the reaction to water. **(An) (1 Mark)**

20. Assertion (A): Magnetic field lines around a bar magnet never intersect each other.

Reason (R): Magnetic field produced by a bar magnet is a quantity that has both magnitude and direction. **(Un) (1 Mark)**

SECTION-B

Questions no. 21 to 26 are Very Short Answer Type questions.

21. (A) List the possible sources of energy required in decomposition reactions. Illustrate any one with a suitable example. **(Re) (2 Marks)**

OR

(B) What is observed when hydrated ferrous sulphate crystals are heated in a dry boiling tube? Give balanced chemical equation (s) of the reaction(s) that occur (s). **(Un) (2 Marks)**

22. Giving reason, state the advantage of using baking powder over baking soda for the preparation of bread or cakes. **(Un) (2 Marks)**

23. State the main function of arteries. Why do they have thick and elastic walls? **(Re, Un) (2 Marks)**

24. (A) Explain how the proteins control the 'characteristics' in an organism with the help of an example of 'short height' trait in pea plant.

(B) Name the information source of making proteins in a cell. **(Un, Re) (2 Marks)**

25. (A) A student has difficulty in reading his textbooks but can read the blackboard clearly while sitting in the last row. Name the defect of vision the student is suffering from. List two reasons due to which this defect arises. Write the nature of the lenses required to correct this defect. **(An) (2 Marks)**

OR

(B) Draw a ray diagram to show the path of a ray of light falling obliquely on one of the refracting faces of a triangular glass prism and mark the angle of deviation on it. **(Cr) (2 Marks)**

26. A voltage source sends a current of 2 A to a resistor of 40Ω connected across it for 5 minutes. Calculate the electrical energy supplied by the source. **(Ev) (2 Marks)**

SECTION-C

Questions no. 27 to 33 are Short Answer Type questions.

27. How is a double displacement reaction different from a displacement reaction? Explain giving example in the form of balanced chemical equations. **(Re) (3 Marks)**

28. (A) Common salt is an important raw material for various chemicals of daily use. State in brief the method of preparation of (i) Sodium hydroxide, and (ii) Sodium hydrogen carbonate from common salt. Write balanced chemical equations of the reactions that occur. **(Ap) (3 Marks)**

OR

(B) Design an experimental set-up to demonstrate that "Alcohol and glucose contain hydrogen but are not categorised as acids". Also give the reason to justify this fact. **(An) (3 Marks)**

29. (A) Enlist any two nitrogenous waste products removed from the blood of human kidney.

(B) Name the capillary cluster formed by the branch of renal artery in the Bowman's capsule.

(C) Depict in the form of a flow chart the path of the urine formed in each kidney until it is finally passed out through the urethra. **(Re, Cr) (3 Marks)**

30. A pure pea plant having round (R), yellow (Y) seeds is crossed with another pure pea plant having wrinkled (r), green (y) seeds. Subsequently F_1 progeny is self-pollinated to obtain F_2 progeny. **(An) (3 Marks)**

(A) What do the seeds of F_1 generation look like?

(B) Give the possible combinations of traits in seeds of F_2 generation. Also give their ratio.

(C) State the reason of obtaining seeds of new combination of traits in F_2 generation.

31. A person has to keep reading material much beyond 25 cm (say at 50 cm) from the eye for comfortable reading. Name the defect of vision he is suffering from. List two causes responsible for arising of this defect. Draw labelled diagram showing correction of this defect using eye-glasses. Are these glasses convergent or divergent of light? **(Cr) (3 Marks)**

32. (A) Draw a labelled diagram to show the pattern of magnetic field lines on a horizontal white board due to a straight current carrying conductor passing perpendicular through its centre. If the direction of current in the conductor is vertically downwards, mark the direction of (i) current, and (ii) magnetic field lines.

- (B) State the right hand thumb rule and check whether the directions marked on the diagram are in accordance with this rule. **(Cr, Re) (3 Marks)**

33. (A) "In a food chain energy flow is unidirectional." Give two reasons for the given statement.
 (B) If 10,000 J energy is available at the producer level, how much energy will be available to the secondary consumers ? Give reason to justify your answer.

(Un, Ap) (3 Marks)

SECTION-D

Questions no. 34 to 36 are Long Answer Type questions.

34. (A) (i) "In refraction of light through a rectangular glass slab, the emergent ray is always parallel to the direction of the incident ray". Why? Explain with the help of a ray diagram. What happens when a ray of light falls normally on one of the faces of a rectangular glass prism? Draw diagram.

- (ii) An object is placed at a distance of 30 cm from the optical centre of a concave lens of focal length 20 cm. Use Lens formula to determine the position of the image formed in this case. **(Cr, Ev) (5 Marks)**

OR

- (B) (i) A student wishes to study the image formation by a concave mirror using candle flame as object. State the type of the image formed by the mirror and mention the change in the image formed, if any, that he observes when the candle flame is gradually moved away from the pole of the mirror. Draw a ray diagram to show the image formation when the object distance is nearly equal to the radius of curvature of the mirror.

- (ii) A convex mirror used for rear-view on an automobile has a focal length of 3.0 m. If a bus is located at 6.0 m from this mirror, use mirror formula to find the position of the image of the bus as seen in the mirror. **(Cr, Ev) (5 Marks)**

35. (A) (i) Consider the following metals:

K, Ca, Al, Cu, Ag, Fe

Select from the above metals, a metal which

- I. does not react with oxygen even at high temperature.
- II. reacts with oxygen at ordinary temperature and forms a protective oxide layer which prevents the metal from further oxidation.
- III. catches fire when kept in the open.
- IV. does not burn in oxygen but the hot metal is coated with a black coloured oxide layer.

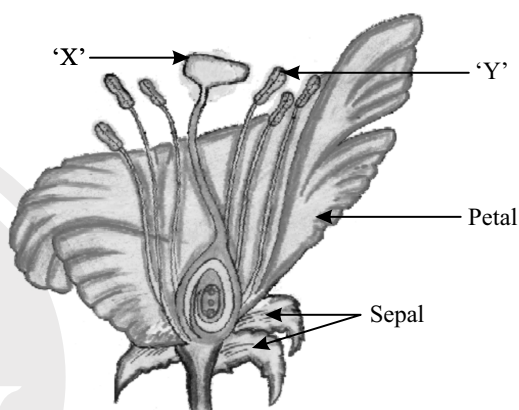
- (ii) What are amphoteric oxides? With the help of balanced chemical equations show that aluminium oxide is an amphoteric oxide.

- (iii) What are alkalis? Give one example. **(Un) (5 Marks)**

OR

- (B) (i) With the help of balanced chemical equations state the process of extracting (I) mercury from its ore called cinnabar. and (II) copper from its sulphide ore.
 (ii) Silver and copper articles slowly lose their shiny surfaces when exposed to air. Name the compounds formed on (I) silver articles, and (II) copper articles in the form of coating. **(Un) (5 Marks)**

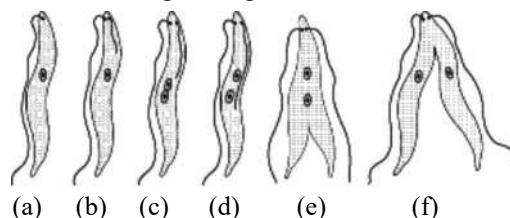
36. (A) (i) Identify the parts 'X' and 'Y' in the figure given below:



- (ii) Name the yellowish coloured structures produced by the part labelled as 'Y'.
 (iii) Write the name of the process by which these are transferred to the part labelled as 'X'.
 (iv) Explain the process of seed formation in a flowering plant. **(Re, Un) (5 Marks)**

OR

- (B) (i) Name the type of asexual mode of reproduction shown in the given figure.



- (ii) Identify the unicellular organism in the diagram.
 (iii) List any two advantages of asexual reproduction over sexual reproduction.
 (iv) Name and explain any one mode of asexual reproduction observed in *Hydra*.

(Re, Un) (5 Marks)

SECTION-E

The following questions are Source-based/Case-based questions. Read the case carefully and answer the questions that follow.

37. 'A' and 'B' are two salts used for washing purposes. Salt 'A' is used for bathing also. Four test tubes I, II, III and IV as mentioned below are taken. (An)

- I. Rain water + solution of salt 'A'
- II. Rain water + solution of salt 'B'
- III. Tubewell water + solution of salt 'A'
- IV. Tubewell water + solution of salt 'B'

The test tubes are shaken one by one almost identically for the same time and the lengths of foam formed in each test tube is noted.

- (i) In which one of the four test tubes is the foam formed the minimum? (1 Mark)
- (ii) Differentiate between salt A and salt B. (1 Mark)
- (iii)(A) What are esters? What happens when an ester reacts with an alkali (say sodium hydroxide)? Give chemical equation for the reaction. (2 Marks)

Or

- (iii)(B) What is the cause of hardness of water? Sometimes it is observed that while bathing foam is formed with difficulty and an insoluble substance is formed. Name this substance and write the cause of its formation. (2 Marks)

38. A person while climbing up a rocky hill comes into a panic state and fear. His body starts reacting in a "flight-or-fight" condition to adjust to the dangerous and stressful situation. (An)

Based on the above facts, answer the questions that follow.

- (i) (A) Name the hormone secreted in the blood of the person in this situation. (2 Marks)

Or

- (i) (B) Name the source gland of the hormone secreted in this condition. (2 Marks)
- (ii) State any two responses in the body of the person as a result of the secretion of this hormone. (1 Mark)

- (iii) How does the action of the chemical signal in terms of hormones differ from the electrical impulses *via* nerve cells? (1 Mark)

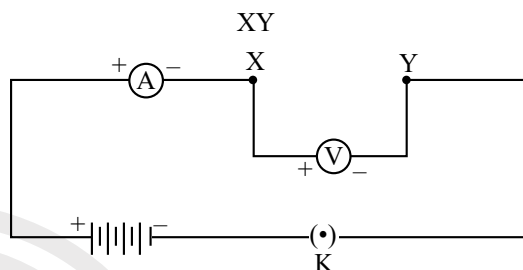
39. As shown in the diagram, an electric circuit consisting of an ammeter, a voltmeter, 4 cells of 1.5 V each, a plug key with a gap XY was set up. Voltmeter and ammeter readings were recorded in the observation table for four arrangements as given below: (An)

Arrangement No. 1- only resistor R_1 in gap XY

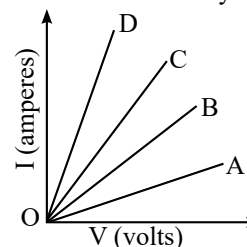
Arrangement No. 2- only resistor R_2 in gap XY

Arrangement No. 3- Resistors R_1 and R_2 in series in gap XY

Arrangement No. 4- Resistors R_1 and R_2 in parallel in gap XY



Based on the observations, four V-I graphs A, B, C and D as shown in figure were drawn. Study these graphs.



- (i) Which one of the graphs represents the series combination of R_1 and R_2 ? (1 Mark)
- (ii) Which one of these graphs represents the parallel combination of R_1 and R_2 ? (1 Mark)
- (iii)(A) Show an arrangement of three resistors, each of resistance $10\ \Omega$, so that the combination has a resistance of $15\ \Omega$. Give justification for your answer. (2 Marks)

Or

- (iii)(B) A battery of 6 V is connected with a series combination of five resistors of $0.1\ \Omega$, $0.2\ \Omega$, $0.3\ \Omega$, $0.4\ \Omega$ and $0.5\ \Omega$. How much current would flow through the $0.3\ \Omega$ resistor? Justify your answer. (2 Marks)

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EXPLANATIONS

1. (c) (ii) and (iii) are isomers since they have the same molecular formula of C_4H_{10} . Hence, they both are related to each other. **(1 Mark)** [NCERT, Ch-4, Pg. no. 65]
2. (d) Hydrogenation of vegetable oils is an addition reaction that occurs using a nickel catalyst or palladium catalyst to give saturated hydrocarbons. **(1 Mark)**
[NCERT, Ch-4, Pg. no. 71]
3. (c) The complete balanced equation is:

$$2Al + 3H_2O \rightarrow Al_2O_3 + 3H_2$$
Hence, $p = 2$, $q = 3$, $r = 1$, $s = 3$ **(1 Mark)**
[NCERT, Ch-1, Pg. no. 3]
4. (c) According to reactivity series, only reactions i) and iv) will take place.

$$CuSO_4 + Mg \rightarrow MgSO_4 + Cu$$

$$ZnSO_4 + Ca \rightarrow CaSO_4 + Zn$$
 (1 Mark)
[NCERT, Ch-3, Pg. no. 44, 45]
5. (b) Salts having the same positive or negative radicals are said to belong to a family. K_2SO_4 , Na_2SO_4 , and $CaSO_4$ belong to the family of sulphate salts. **(1 Mark)**
[NCERT, Ch-2, Pg. no. 28, 29]
6. (d) Fertilisation of egg and sperm occurs in the oviduct, embryo implants in the uterus, sperm enters through the vagina, and the placenta removes waste from the developing embryo's blood, ensuring proper nutrient and waste exchange. **(1 Mark)**
[NCERT, Ch-7, Pg. no. 124]
7. (d) The medulla in the hindbrain controls involuntary actions like salivation and vomiting, regulating essential reflexes for body function. **(1 Mark)**
[NCERT, Ch-6, Pg. no. 104]
8. (b) *Hibiscus* and Mustard are bisexual flowers, containing both stamens (male) and carpels (female), enabling self and cross-pollination. **(1 Mark)**
[NCERT, Ch-7, Pg. no. 120]
9. (b) Cytokinins promote cell division and are found in high concentrations in actively growing regions like fruits and seeds. **(1 Mark)** [NCERT, Ch-6, Pg. no. 108]
10. (b) *Amoeba* forms pseudopodia, a temporary finger-like extensions, to engulf food through phagocytosis, aiding movement and nutrition intake. **(1 Mark)**
[NCERT, Ch-5, Pg. no. 84]
11. (c) The resistance R of a wire is given by $R = \rho \frac{l}{A}$
When the wire is stretched to 3 times its original length ($3l$), the volume of the wire remains the same, so

$$l \times A = (3l) \times A' \Rightarrow A' = \frac{A}{3}$$

Thus, the new resistance R' becomes

$$R' = \rho \frac{3l}{A/3} = \rho \frac{3l \times 3}{A} = 9\rho \frac{l}{A} = 9R$$

(1 Mark) [NCERT, Ch-11, Pg. no. 178]

12. (d)

- Outside the magnet, magnetic field lines run from the north pole to the south pole.
- Inside the magnet, they run from the south pole to the north pole to form closed loops. **(1 Mark)**

[NCERT, Ch-12, Pg. no. 197]

13. (a) The absolute refractive index n of a medium is given by

$$n = \frac{c}{v}$$

For glass:

$$n_g = \frac{c}{v_g}$$

Substituting values:

$$\frac{3}{2} = \frac{c}{2 \times 10^8}$$

Solving for c :

$$c = \frac{3}{2} \times 2 \times 10^8 = 3 \times 10^8 \text{ m/s}$$

For water

Using the same formula:

$$n_w = \frac{c}{v_w}$$

Substituting values:

$$\frac{4}{3} = \frac{3 \times 10^8}{v_w}$$

Solving for v_w :

$$v_w = \frac{3 \times 10^8}{4/3} = 3 \times 10^8 \times \frac{3}{4} = \frac{9}{4} \times 10^8 \text{ m/s}$$

(1 Mark) [NCERT, Ch-9, Pg. no. 143]

14. (c) When a beam of white light passes through fine dust particles, the color that scatters the most is blue.

This phenomenon is called Rayleigh scattering, where shorter wavelengths of light scatter more readily, and blue light has the shortest wavelength in the visible spectrum.

(1 Mark) [NCERT, Ch-10, Pg. no. 169]

15. (c) While the direction of the current determines the polarity of the magnetic field (north or south pole), it does not affect the strength of the field inside the solenoid. **(1 Mark)** [NCERT, Ch-12, Pg. no. 201]

16. (d) A self-sustaining aquarium with herbivorous fish needs aquatic plants for oxygen and food and decomposers for nutrient cycling. Terrestrial organisms and specific consumers like clownfish and sea urchins are unnecessary.

(1 Mark) [NCERT, Ch-13, Pg. no. 208, 209]

17. (b) The nervous system controls voluntary actions by sending signals to muscles. While Reason (R) mentions the PNS components, it does not explain how the CNS communicates with the body through the PNS. (1 Mark) [NCERT, Ch-6, Pg. no. 103]

18. (a) Using jute bags reduces pollution because jute is biodegradable and reusable, unlike plastic bags that contribute to environmental pollution. Jute bags decompose naturally and can be used multiple times, making them an eco-friendly alternative to plastic.

(1 Mark) [NCERT, Ch-13, Pg. no. 214]

19. (c) Hydrogen gas is not evolved when a metal reacts with nitric acid. It is because HNO_3 is a strong oxidising agent. It oxidises the H_2 produced to water and itself gets reduced to any of the nitrogen oxides (N_2O , NO , NO_2).

(1 Mark) [NCERT, Ch-3, Pg. no. 44]

20. (b) Both A and R are true but R is not the correct explanation for A.

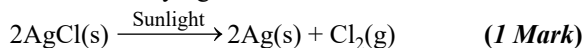
The assertion is correct because magnetic field lines, by their very nature, indicate the direction of the magnetic field at every point, and if they were to intersect, it would imply the existence of two different directions at the same point, which is impossible. Although the reason is true (magnetic fields are vector quantities with both magnitude and direction), it does not explain why the field lines never intersect. The non-intersecting property is due to the uniqueness of the magnetic field vector at any given point rather than simply because it is a vector quantity. (1 Mark)

[NCERT, Ch-12, Pg. no. 197]

21. (A) Decomposition reactions require an external source of energy to break down compounds into simpler substances. The common sources of energy include:

- (1) Heat (Thermal Decomposition)
- (2) Light (Photodecomposition)
- (3) Electricity (Electrolytic Decomposition) (1 Mark)

Example: Decomposition of silver chloride into silver and chlorine by light.

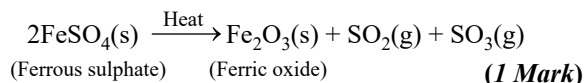


[NCERT, Ch-1, Pg. no. 8-10]

OR

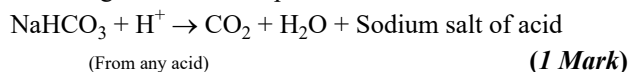
- (B) Ferrous sulphate crystals ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) lose water when heated and the colour of the crystals changes. It then decomposes to ferric oxide (Fe_2O_3), sulphur dioxide (SO_2) and sulphur trioxide (SO_3) (1 Mark)

The balanced chemical equation is:



[NCERT, Ch-1, Pg. no. 8]

22. Baking powder is a mixture of baking soda (sodium hydrogencarbonate) and a mild edible acid such as tartaric acid. When baking powder is heated or mixed in water, the following reaction takes place:



[NCERT, Ch-2, Pg. no. 31]

Carbon dioxide produced during the reaction can cause bread or cake to rise making them soft and spongy. Thus, baking powder is used over baking soda in bakery.

(1 Mark)

- 23.

- Main function of arteries: Arteries carry oxygenated blood from the heart to different parts of the body.

(1 Mark)

- Arteries carry blood under high pressure from the heart. To sustain this force, they have thick, elastic walls that prevent rupture and maintain continuous blood flow.

(1 Mark)

[NCERT, Ch-5, Pg. no. 93]

24. (A) Proteins control traits by regulating hormone production. (½ Mark)

Example: In pea plants if a gene controls height by producing an enzyme that regulates growth hormone. A functional enzyme results in more hormone, making the plant tall, while a less efficient enzyme produces less hormone, leading to short plants. (½ Mark)

- (B) The DNA in a cell provides the genetic information required for protein synthesis. (1 Mark)

[NCERT, Ch-8, Pg. no. 131]

25. (A) The student is suffering from hypermetropia (farsightedness), where the near point of clear vision is abnormally far from the eye. This defect arises primarily due to: (½ Mark)

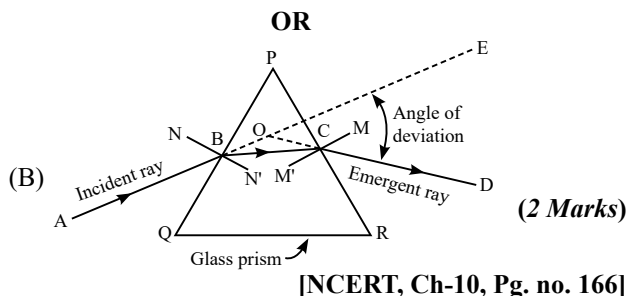
1. An eyeball that is shorter than normal, causing the image of near objects to be focused behind the retina.

(½ Mark)

2. Insufficient refractive power of the cornea or lens, which fails to adequately converge light rays from close objects onto the retina. (½ Mark)

To correct this defect, convex (converging) lenses are used. These lenses increase the overall refractive power of the eye, thereby shifting the image forward onto the retina for clear near vision. (½ Mark)

[NCERT, Ch-10, Pg. no. 163]



26. First, we determine the voltage across the resistor using Ohm's law:

$$V = I \times R = 2 \text{ A} \times 40 \Omega = 80 \text{ V.} \quad (\frac{1}{2} \text{ Mark})$$

The power supplied by the source is:

$$P = V \times I = 80 \text{ V} \times 2 \text{ A} = 160 \text{ W.} \quad (\frac{1}{2} \text{ Mark})$$

Since the time is given in minutes, convert 5 minutes to seconds:

$$t = 5 \text{ min} \times 60 \text{ s/min} = 300 \text{ s.}$$

Thus, the electrical energy supplied is:

$$E = P \times t = 160 \text{ W} \times 300 \text{ s} = 48,000 \text{ J.} \quad (1 \text{ Mark})$$

[NCERT, Ch-11, Pg. no. 191]

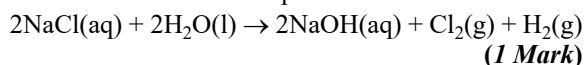
27. Displacement Reaction and Double Displacement Reaction:

Displacement Reaction	Double Displacement Reaction
Involves a single element being replaced in a compound.	Involves exchange of different atoms or groups of atoms (ions).
More reactive element displaces less reactive one.	No element displacement, only ion exchange.
Often observable through a change in the solution colour or deposition.	Often results in the formation of a precipitate.
Forms a new compound and a different element.	Forms two new compounds.
Example: Iron nail in copper sulphate solution. $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$	Example: Reaction of sodium sulphate with barium chloride. $\text{Na}_2\text{SO}_4(\text{aq}) + \text{BaCl}_2(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$

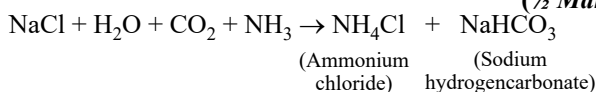
(3 Marks) [NCERT, Ch-1, Pg. no. 10-12]

28. (A) (i) When electricity is passed through an aqueous solution of sodium chloride (called brine), it decomposes to form sodium hydroxide. The process is called the chlor-alkali process. ($\frac{1}{2}$ Mark)

The balanced chemical equation that is involved is:

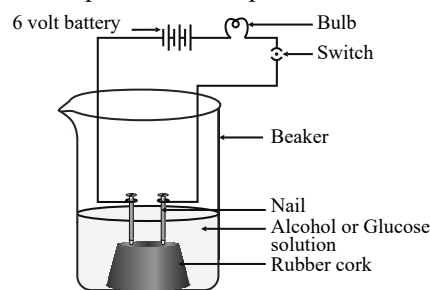


- (ii) Sodium hydrogencarbonate (NaHCO_3) is produced using sodium chloride as shown below: ($\frac{1}{2}$ Mark)



(1 Mark) [NCERT, Ch-2, Pg. no. 30, 31]

- OR**
- (B) The experimental setup is shown below:



(1 Mark)

In this experiment, the bulb not glowing in the presence of glucose and alcohol solutions suggests that these substances do not conduct electricity. This is because they do not dissociate into ions in solution. (1 Mark)

Even though both glucose and alcohol contain hydrogen atoms, they do not release free ions (such as H^+ or OH^-) in water, which are required for electrical conductivity. Thus, these solutions cannot be classified as acids. (1 Mark)

[NCERT, Ch-2, Pg. no. 22, 23]

29. (A) Two nitrogenous waste products removed by the kidney

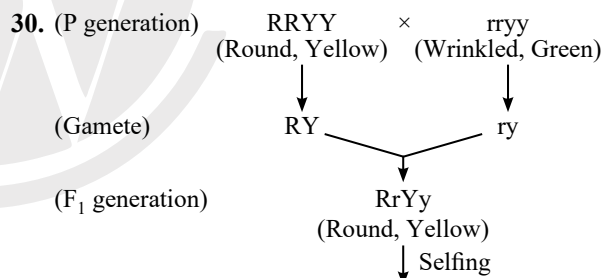
- Urea ($\frac{1}{2}$ Mark)
- Uric acid ($\frac{1}{2}$ Mark)

- (B) The glomerulus is the capillary cluster formed by the branch of the renal artery in Bowman's capsule. (1 Mark)

- (C) Flowchart depicting the path of urine formation and excretion

Kidney $\rightarrow$ Ureter $\rightarrow$ Urinary Bladder $\rightarrow$ Urethra $\rightarrow$
Urine expelled from the body (1 Mark)

[NCERT, Ch-5, Pg. no. 96, 97]



	RY	Ry	rY	ry
RY	RRYY (Round Yellow)	RRYy (Round Yellow)	RrYY (Round Yellow)	RrYy (Round Yellow)
Ry	RRYy (Round Yellow)	RRyy (Round Green)	RrYy (Round Yellow)	Rryy (Round Green)
rY	RrYY (Round Yellow)	RrYy (Round Yellow)	rrYY (Wrinkled Yellow)	rrYy (Wrinkled Yellow)
ry	RrYy (Round Yellow)	Rryy (Round Green)	rrYy (Wrinkled Yellow)	rryy (Wrinkled Green)

- (A) The F₁ generation seeds will all be round and yellow because round (R) and yellow (Y) are dominant traits over wrinkled (r) and green (y). (1 Mark)

(B)

Possible combination of traits in F ₂ generation	Genotypic combination	Ratio
Round, Yellow	RRYY, RRYy, RrYY, RrYy	9/16
Round, Green	RRyy, Rryy	3/16
Wrinkled, Yellow	rrYY, rrYy	3/16
Wrinkled, Green	rryy	1/16

($\frac{1}{4} \times 4 = 1$ Mark)

(C) New combinations arise due to the independent assortment of genes during gamete formation.

($\frac{1}{2}$ Mark)

The alleles for seed shape (R, r) and seed color (Y, y) segregate independently, leading to new trait combinations.

($\frac{1}{2}$ Mark)

[NCERT, Ch-8, Pg. no. 131]

31. The defect of vision that the person is suffering from is called 'Hyperopia' or 'Farsightedness'. This condition occurs when the eye is too short or the cornea has too little curvature, causing light to focus behind the retina instead of directly on it. As a result, the person can see distant objects clearly but struggles to see nearby objects.

(1 Mark)

Two causes of hyperopia are:

- **Shortened eyeball**

The eyeball becomes too small in length, which reduces the distance between the retina and the eye lens.

($\frac{1}{2}$ Mark)

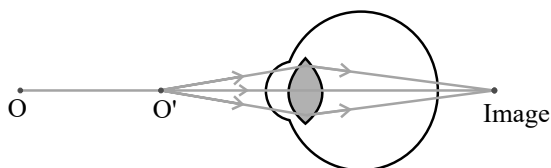
- **Elongated eye lens**

The focal length of the eye lens becomes too long, which reduces the eye lens's ability to converge.

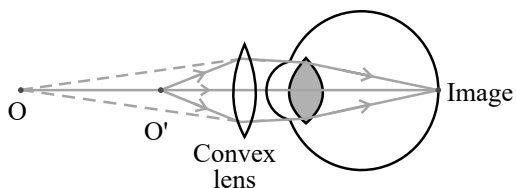
($\frac{1}{2}$ Mark)

To correct hyperopia, convex lenses (converging lenses) are used in eyeglasses. These lenses help to bend the light rays inward so that they focus directly on the retina, allowing for clear vision at close distances.

Hypermetropic eye:



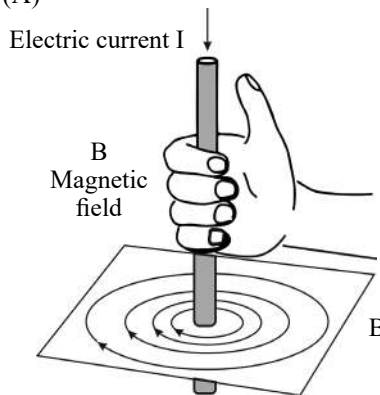
Correction using convex lens



($\frac{1}{2}$ Mark)

[NCERT, Ch-31, Pg. no. 163]

32. (A)



($\frac{1}{2}$ Mark)

(B) **Right Hand Thumb Rule:**

Statement: Point the thumb of the right hand in the direction of the current (here, into the board). Then, the curl of the fingers gives the direction of the magnetic field lines.

(1 Mark)

Verification: With the thumb pointing downward (into the board), the fingers naturally curl in a clockwise direction when seen from above. This confirms that the magnetic field lines in the diagram are correctly drawn as clockwise, in accordance with the right hand thumb rule.

($\frac{1}{2}$ Mark) [NCERT, Ch-12, Pg. no. 200]

33. (A) Energy flow in a food chain is unidirectional because:

- One-way transfer of energy: Energy captured by autotrophs doesn't return to the sun, and energy passed to herbivores doesn't revert to autotrophs.

($\frac{1}{2}$ Mark)

- Energy loss at each step: Due to the 10% law, most energy is lost as heat, limiting its transfer to the next level.

($\frac{1}{2}$ Mark)

(B) Energy at producer level = 10,000 J

Using the 10% Law which states that only 10% of energy is passed from one trophic level to the next, while 90% is lost as heat.

(1 Mark)

- Primary consumers receive:

$$10\% \text{ of } 10,000 \text{ J} = 1,000 \text{ J}$$

($\frac{1}{2}$ Mark)

- Secondary consumers receive:

$$10\% \text{ of } 1,000 \text{ J} = 100 \text{ J}$$

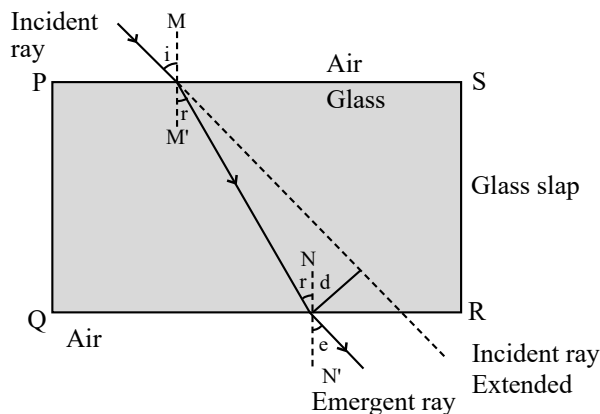
($\frac{1}{2}$ Mark)

Thus, 100 J of energy will be available to the secondary consumers. [NCERT, Ch-13, Pg. no. 210, 211]

34. (A) (i) **Refraction Through a Rectangular Glass Slab:**

When a ray enters a glass slab at an angle, it refracts toward the normal because the speed of light decreases in glass. On reaching the second surface, the ray goes from glass to air and refracts away from the normal. The geometry of the slab is such that the second refraction cancels the angular deviation produced at the first surface. Thus, the emergent ray is parallel to the incident ray, even though it is laterally shifted.

(1 Mark)

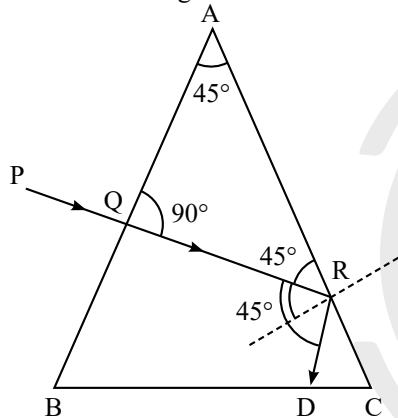


(½ Mark)

Normal Incidence on a Rectangular Glass Prism:

When the ray falls perpendicularly (normal incidence) on one face, the angle of incidence is 0° . According to Snell's law, no refraction occurs because $\sin 0^\circ = 0$; hence, the ray travels straight through without bending. There may be a lateral shift, but the direction remains unchanged.

(1 Mark)



(½ Mark)

(ii) Using the lens formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Here, for a concave lens, the focal length f is negative, and by sign convention, the object distance u is also taken as negative. Thus,

$$f = -20 \text{ cm}$$

$$u = -30 \text{ cm}$$

Substitute into the lens formula:

$$\frac{1}{-20} = \frac{1}{v} + \frac{1}{-30} \Rightarrow \frac{1}{-20} = \frac{1}{v} - \frac{1}{30}$$

Rearrange to solve for $\frac{1}{v}$:

$$\frac{1}{v} = -\frac{1}{20} + \frac{1}{30}$$

$$\frac{1}{v} = -\frac{3}{60} + \frac{2}{60} = -\frac{1}{60}$$

Thus, $v = -60 \text{ cm}$

The negative sign indicates that the image is virtual and formed on the same side of the lens as the object.

(2 Marks) [NCERT, Ch-9, Pg. no. 148]

OR

(B) (i) A concave mirror forms a **real, inverted image** of a candle flame when the flame (object) is placed beyond the focal point. As the candle is gradually moved away from the pole (i.e. increasing the object distance), the image changes as follows: (1 Mark)

- When the object is between the focus (F) and the centre of curvature (C), the image is **real, inverted, and magnified** and appears beyond C.
- When the object is exactly at C (i.e. when the object distance $\approx$ radius of curvature), the image is real, inverted, of the same size as the object and is formed at C.
- For further increases in object distance (object beyond C), the image remains real and inverted but becomes **diminished and moves closer to F**. (2 Marks)

[NCERT, Ch-9, Pg. no. 140]

(ii) Using the mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

where

$$u = -6 \text{ m}$$

$$f = 3 \text{ m}$$

putting in mirror formula

$$\frac{1}{3} = \frac{1}{v} + \frac{1}{-6}$$

$$\frac{1}{3} + \frac{1}{6} = \frac{1}{v}$$

$$\frac{6+3}{18} = \frac{1}{v}$$

$$\frac{9}{18} = \frac{1}{v}$$

$$v = 2 \text{ m} \quad (2 \text{ Marks}) \quad [\text{NCERT, Ch-9, Pg. no. 143}]$$

35. (A) (i)

I: Silver (Ag) does not react with oxygen even at high temperature. (½ Mark)

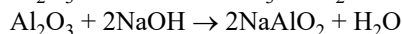
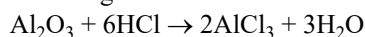
II: Al reacts with oxygen at ordinary temperature and forms a protective oxide layer which prevents the metal from further oxidation. (½ Mark)

III: Potassium catches fire when kept in open. (½ Mark)

IV: Copper does not burn, but the hot metal is coated with a black coloured layer of copper(II) oxide. (½ Mark)

(ii) Metal oxides which react with both acids as well as bases to produce salts and water are known as amphoteric oxides. (1 Mark)

Aluminium oxide reacts with acids and bases in the following manner:



(Sodium aluminate) (1 Mark)

- (iii) An alkali is a base that dissolves in water. Most metal oxides are insoluble in water but some of these dissolve in water to form alkalis. (½ Mark)

Example: Sodium oxide dissolve in water to produce sodium hydroxide which is an alkali.

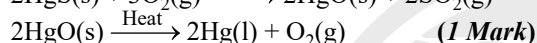
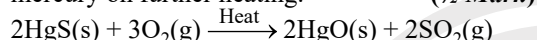


[NCERT, Ch-3, Pg. no. 41, 42]

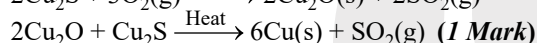
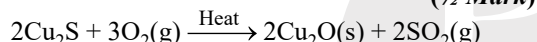
OR

- (B) (i)

- (I) Cinnabar (HgS) is an ore of mercury. When it is heated in air, it is first converted into mercuric oxide (HgO). Mercuric oxide is then reduced to mercury on further heating. (½ Mark)



- (II) Copper which is found as Cu_2S in nature can be obtained from its ore by just heating in air. (½ Mark)



- (ii)

- (I) Silver sulphide, Ag_2S (1 Mark)

- (II) Basic Copper carbonate, $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$ (1 Mark)

[NCERT, Ch-3, Pg. no. 51-54]

36. (A) (i)

- $\text{X} \rightarrow$ Stigma (½ Mark)
- $\text{Y} \rightarrow$ Anther (½ Mark)

- (ii) The anther (Y) produces pollen grains, which are yellow-colored and contain male gametes for fertilization. (1 Mark)

- (iii) The process by which pollen grains are transferred to the stigma is called pollination. (1 Mark)

- (iv) Process of seed formation in a flowering plant:

- Pollination – Transfer of pollen grains from the anther to the stigma. (½ Mark)
- Fertilization – Pollen reaches the ovule in the ovary, where the male gamete fuses with the female gamete to form a zygote. (½ Mark)
- Embryo Formation – The zygote divides repeatedly to form an embryo inside the ovule. (½ Mark)
- Seed and Fruit Formation – The ovule develops into a seed, while the ovary ripens into a fruit, which protects the seed. (½ Mark)

[NCERT, Ch-7, Pg. no. 120-121]

OR

- (B) (i) The diagram shows binary fission. (1 Mark)

- (ii) The organism shown is *Leishmania*. (1 Mark)

- (iii) Two advantages of asexual reproduction over sexual reproduction

1. Faster Reproduction – Asexual reproduction is rapid, allowing quick population growth without requiring a mate. (½ Mark)
2. Genetic Stability – Offspring are genetically identical to the parent, ensuring stability of traits in favorable conditions. (½ Mark)

- (iv) *Hydra* reproduces asexually by budding. (1 Mark)

A small outgrowth forms due to repeated cell division, gradually enlarges, and develops tentacles. Once fully matured, it detaches from the parent and becomes an independent organism, allowing rapid reproduction.

(1 Mark) [NCERT, Ch-7, Pg. no. 115-117]

37. (i) The foam formation is the minimum in **Test tube III (Tubewell water + solution of salt 'A')**. (½ Mark)

This is because **Salt A is used for bathing purposes** and is likely to be a soap and tubewell water contains salts of **magnesium and calcium**, which cause water hardness and prevent foam formation. (½ Mark)

- (ii)

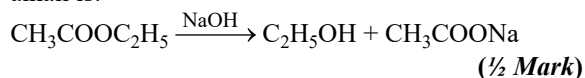
Salt A	Salt B
Used for both bathing and washing purposes.	Used only for washing purposes.
Likely to be a soap (sodium or potassium salt of fatty acid).	Likely to be a detergent (synthetic cleaning agent that works in hard water).
Does not form foam in hard (tubewell) water due to the presence of calcium and magnesium salts.	Forms foam easily in both rainwater and tubewell water

(1 Mark)

- (iii) (A) Esters are sweet-smelling substances formed by reaction of an acid and an alcohol. (1 Mark)

On treating with sodium hydroxide, which is an alkali, the ester is converted back to alcohol and sodium salt of carboxylic acid. (½ Mark)

The chemical equation for the reaction of ester with an alkali is:



Or

- (iii) (B) Hardness of water is due to the presence of calcium and magnesium salts. **(1 Mark)**

While bathing, foam is formed with difficulty due to the formation of the insoluble substance called scum which remains after washing with water. **(½ Mark)**

This is caused by the reaction of soap with the calcium and magnesium salts, which cause the hardness of water. **(½ Mark)**

[NCERT, Ch-4, Pg. no. 73-76]

38. (i) (A) The hormone secreted in this situation is Adrenaline. **(1 Mark)**

It is released in response to stress or fear, triggering the fight-or-flight response. **(1 Mark)**

Or

- (i) (B) The hormone Adrenaline is secreted by a pair of glands called Adrenal Glands. **(1 Mark)**

These glands are located above the kidneys. **(1 Mark)**

- (ii) Two responses in the body due to adrenaline secretion:

- The heart beats faster, supplying more oxygen to muscles for immediate action. **(½ Mark)**
- The diaphragm and rib muscles contract, allowing faster oxygen intake, supporting the body's fight-or-flight response. **(½ Mark)**

- (iii) Hormonal action is slow, traveling through blood to affect multiple organs, while nerve impulses are fast, transmitting electrical signals for immediate responses. **(1 Mark)**

[NCERT, Ch-6, Pg. no. 108, 109]

39. (i) In a series circuit, the total resistance increases, resulting in a lower current for the same voltage.

The V-I graph with the least slope (smallest current) corresponds to the highest resistance.

Hence, graph A represents the series combination of R_1 and R_2 . **(1 Mark)**

- (ii) In a parallel circuit, the total resistance decreases, leading to a higher current for the same voltage.

The V-I graph with the steepest slope (highest current) corresponds to the lowest resistance. **(1 Mark)**

Hence, **graph D** represents the parallel combination of R_1 and R_2 .

- (iii) (A) We can combine two resistors in parallel and then add the third in series:

Parallel combination of two $10\ \Omega$ resistors:

$$R_p = \frac{10 \times 10}{10 + 10} = 5\ \Omega$$

Adding a third $10\ \Omega$ resistor in series

$$R_{eq} = 5 + 10 = 15\ \Omega$$

Or

- (iii) (B) Total resistance of the series circuit:

$$R_{total} = 0.1 + 0.2 + 0.3 + 0.4 + 0.5 \\ = 1.5\ \Omega$$

Total current using Ohm's law:

$$I = \frac{V}{R} = \frac{6\text{ V}}{1.5\ \Omega} = 4\text{ A}$$

Since it is a series circuit, the same 4A current flow through all resistors, including the $0.3\ \Omega$ resistor.

(2 Marks) [NCERT, Ch-11, Pg. no. 176]

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